

Tutorial 3

1. Standard ACL - Refer to T4 ACL Workbookv2:

- Page 5
- Page 17
- Page 26-34 (problem 1,3,5, 9)

2. Extended ACL - Refer to T4 ACL Workbookv2:

- Page 9
- Page 38(problem 1)
- Page 44 (problem 5)
- Page 50 (problem 9)
- Page 60 (problem 15)
- Page 64 (problem 19).

3. **Access Control Lists (ACLs)** are used to filter network traffic. Based on **Figure 1-1**, answer the following questions. Assume all hosts and routers can communicate with each other.

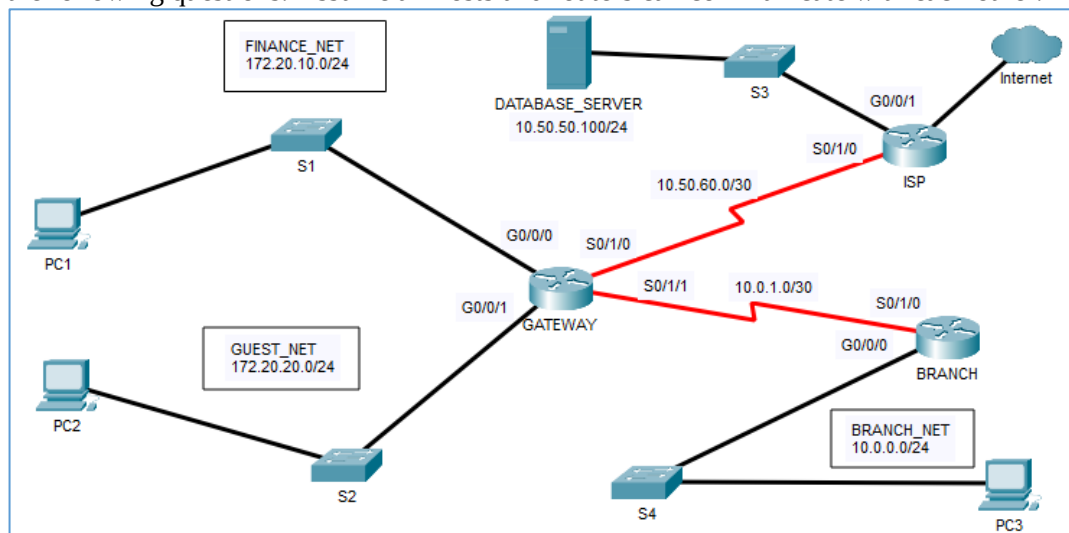


Figure 1-1: A network topology

- Write an **Extended Named ACL** called **SECURE_ACCESS** to allow only **even-numbered** host IP addresses from **FINANCE_NET** network to access the **DATABASE_SERVER** via **SQL** (Port 1433). Allow **GUEST_NET** network to access the Internet via **HTTPS** (443) but deny them access any internal **BRANCH_NET** network addresses. Explicitly deny all other traffic. Use port number for services and suitable keyword(s) in your ACL. Specify the router, interface(s) and direction for the most efficient placement. (17 marks)
- Explain the concept of "**Implicit Deny**" in ACLs and how it affects traffic if a "**permit any**" statement is omitted. (4 marks)

4. Differentiate the following features for a standard ACL and an extended ACL

Feature	Standard ACL	Extended ACL
Filters (permit or deny) Based On		
Control Level		
ACL Number Range		
Placement		
Protocols Supported		
Port Filtering		
Typical Use		
OSI model		

5. Based on Figure 2-1, illustrate the differences of applying ACL on the incoming port or outgoing port of Router1.

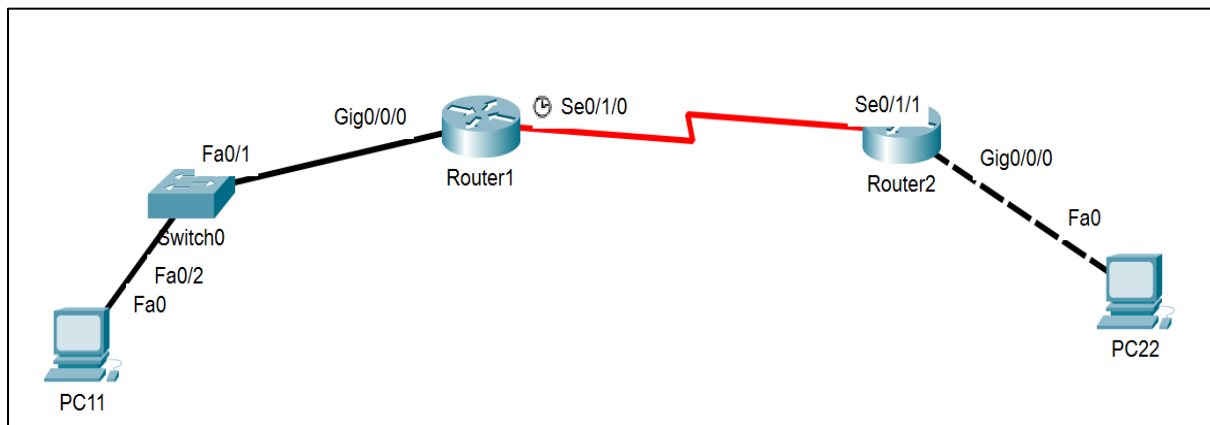


Figure 2-1: A network topology

6. a) With reference to Figure 3-1, you are required to explain the theoretical concepts to your CEO with the following access control entries (ACEs) to filter traffic before reaching 172.50.25.0/24 network.

```
ip access-list standard PERMIT_UPPER
permit host 212.77.7.2
permit 212.77.7.128 0.0.0.127
deny 212.77.7.0 0.0.0.255
permit any
```

```
int g0/0/0
ip access-group Permit_Upper out
```

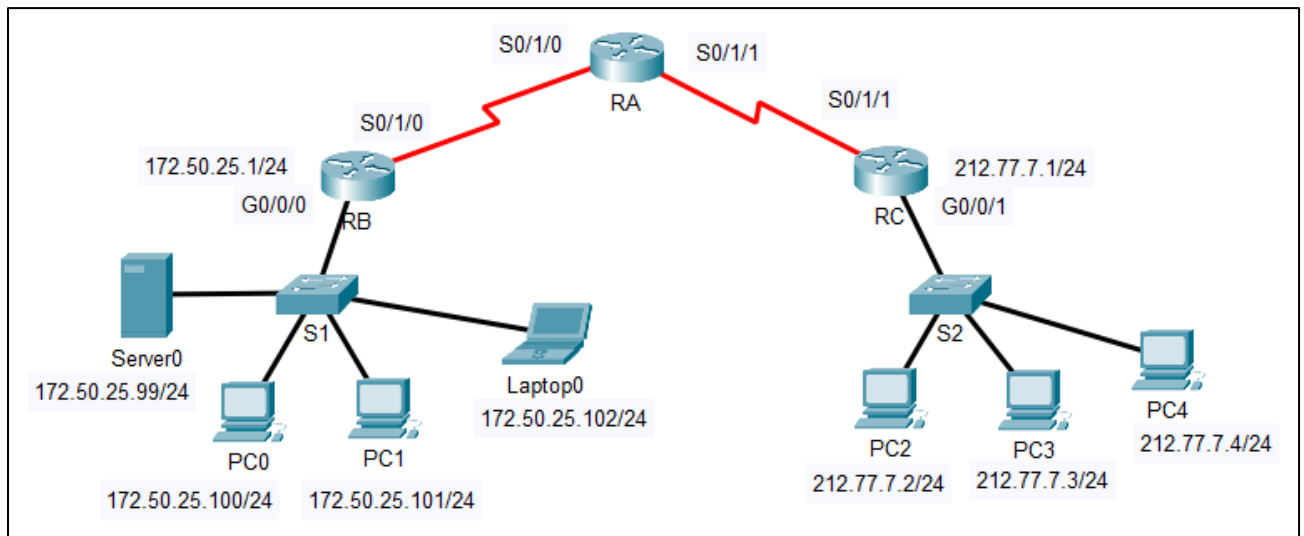


Figure 3-1: A network topology

- b) With reference to Figure 3-1, you are required to explain to your CEO with the following access control entries (ACEs).

```
access-list 185 deny tcp 212.77.7.0 0.0.0.15 host 172.50.25.99 eq www
access-list 185 permit tcp 212.77.7.0 0.0.0.255 host 172.50.25.99 eq www
access-list 185 deny ip any any
```

```
int g0/0/1
ip access-group 185 in
```